EEG-based Emotion Recognition: An In-depth Analysis using DEAP and SEED Datasets

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***Abstract*— Research on emotion recognition has made an increasing amount of emphasis on the understanding of Electroencephalogram (EEG) signals. Using two well-known datasets - the SEED (SEED Dataset for Emotion Analysis using EEG) and the DEAP (Dataset for Emotion Analysis using Physiological Signals), this work explores the complex analysis of EEG signals and their use in emotion recognition. The study highlights important characteristics suggestive of emotional states while delving into the basic ideas behind the acquisition, processing, and interpretation of EEG signals. We explore the viability and efficiency of using EEG signals for emotion recognition tasks by utilizing machine learning and signal processing techniques. We also explore the opportunities and problems associated with EEG-based emotion identification systems, such as feature selection, artefact removal, and signal noise. The goal of this study is to offer researchers and practitioners useful insights into using EEG signals for emotion identification applications through a thorough evaluation and analysis. showcasing the effectiveness of our methodology in EEG-based emotion recognition. The study demonstrates promising arousal accuracy at 70.88% and notable valence accuracy at 76.00% using SVM Classifier.**

Keywords—Electroencephalogram (EEG), signal noise, artefact removal, feature extraction, DEAP and SEED datasets.

I. Introduction

Comprehending human emotions is an enduring pursuit that has captivated scholars from several fields. Understanding emotions is essential for a number of disciplines, including psychology, neuroscience, and human-computer interaction [1], since emotions are fundamental to human behaviour, cognition, and interpersonal interactions. While self-reporting and behavioural observation play a major role in traditional methods of emotion evaluation, recent developments in neuroimaging techniques have offered hitherto untapped insights into the brain underpinnings of emotions. Electroencephalogram (EEG) data have become a potent tool among these methods for studying brain activity related to emotional experiences [2]. Through the use of scalp electrodes shown in below Figure 1, EEG records brain electrical activity and provides a clear insight into the dynamics of the neurological processes that underlie emotions. These electrodes are applied to the scalp in order to assess the voltage variations brought on by ionic current flows in brain neurons. According to the worldwide 10-20 system, which offers a standardised procedure for electrode placement, each electrode is normally labelled. Deep implications for a wide range of applications, such as affective computing, mental health monitoring, and human-computer interaction, result from the capacity to read emotional states from EEG signals.

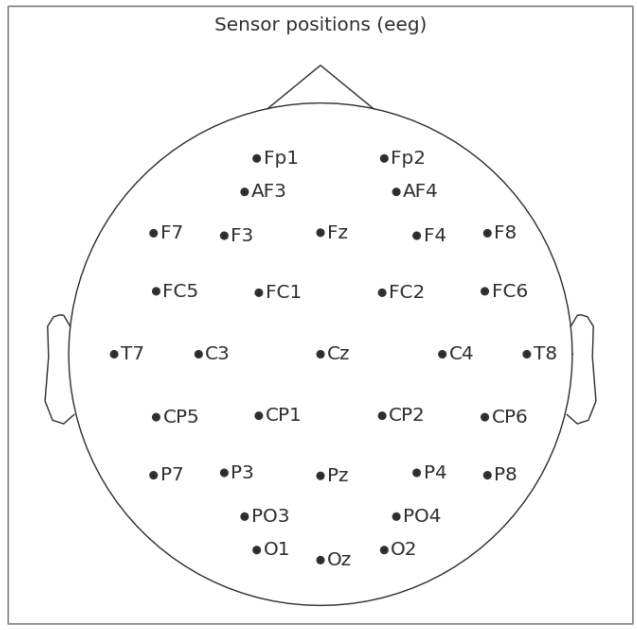

Fig. 1. EEG Scalp Electrode Representation

In Figure 1, frontal, temporal, central, parietal, and occipital regions of the brain are represented by the letters F, T, C, P, and O [3]. The left hemisphere is represented by odd numbers, while the right hemisphere by even ones. The anterior-posterior position is indicated by the numbers (1, 2, 3, 4, 5, 6, 7, 8). For instance, F3 is located more anteriorly than F7 and on the left side of the frontal lobe. Midline position is represented by the letter 'z'. Because these regions are frequently linked to emotional processing, electrodes are usually placed across the frontal (F), central (C), and temporal

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(T) regions for emotion recognition. For instance, frontopolar areas (FP1, FP2). Frontal areas F3, F4, F7, and F8, Central regions C3, C4, Temporal regions T3, T4, T5, and T6, O1, O2 Regions of occipital specific electrodes capture the electrical activity in specific brain regions, which can subsequently be examined to determine the person's emotional state.

Our goal in this research is to clarify the nuances of EEG signals and how they are used in emotion recognition tasks. We concentrate on two well-known datasets: the Sichuan University Emotion EEG Dataset (SEED) and the Dataset for Emotion Analysis using Physiological Signals (DEAP). These datasets include extensive archives of EEG recordings made during a range of emotional stimuli, for in-depth examinations of the brain's reactions in diverse emotional states. Our study includes a detailed analysis of feature extraction strategies, EEG signal processing techniques, and classification algorithms designed for emotion recognition applications. We explore the difficulties in signal artefacts, noise interference, and feature relevance that arise in EEG-based emotion identification. Our goal is to improve the accuracy and resilience of EEG-based emotion recognition systems in order to make it easier for these systems to be integrated into practical applications. The field of emotion recognition from EEG data has seen a revolution in recent years thanks to the incorporation of sophisticated machine learning and deep learning algorithms [4]. Techniques like Multiple Column Convolutional Neural Network (MCCNN) models [5] provide improved capabilities for obtaining hierarchical features from EEG data, which increases task robustness and accuracy for classifying emotions. Additionally, by combining deep learning with mixed-model (Ensemble Techniques) [6] [7] machine learning techniques, more sophisticated and context-aware emotion recognition systems are made possible by leveraging the complimentary advantages of both paradigms. Utilising these methods' benefits—such as their capacity to recognise complex patterns and adjust to a range of emotional situations—holds great potential for improving the state-of-the-art in EEG-based emotion recognition.

II. Literature Review

A method for extracting the features of the data was presented that used dynamic differential probability. CNN was then used to classify the dynamic differential entropy (DDE) features that were collected. To extract the spatial features of fMRI data and obtain spatial information sorted by time, the author first developed a convolutional denoising autoencoder, which combines the benefits of convolutional neural networks (CNNs) with the denoising autoencoder (DAE). The author [8] determined the ideal profile of electrode sites by analysing the brain region that is closely associated with emotions in order to reduce the computational complexity. Experimental data indicates that the model's accuracy is 3.50% more than that of alternative systems [9]. The author [10] set out to use an optimised CNN to create a well-adapted emotion identification model. This work uses an optimised deep convolutional neural network (Deep CNN) to develop an appropriate emotion detection model. This research is significant because of the suggested hybrid hunt optimisation, which modifies the Deep CNN hyper-parameters and selects the informative electrodes based on brain activity. The proposed hybrid hunt optimisation, which includes altering Deep CNN hyper-parameters and selecting pertinent electrodes based on brain activity, is the main contribution of this work. For the two datasets, the accuracy of the suggested recognition model is 96.60% and 95.80%, respectively.

The electrode-specific functions relation of the Attention Mechanism [11] is crucial for the recognition of emotion by EEG multichannel data. In the part that follows, data processing and methods are explained in great detail. It makes sense to combine LSTM and CNN capabilities to create hybrid deep learning models. The authors, in [12] and [13], recommends a wavelet-based pre-processing to turn the multi-channel EEG into a 2D frame representation. Each frame represents a time slice of the multichannel EEG's energy distribution. DGCNN[14] is another significant technique that allows for the extraction of various features and, subsequently, the demonstration of good emotion identification accuracy. Using ensemble approaches, many models are integrated to improve resilience and overall performance. The article describes the use of a multi-source domain transfer discriminative dictionary learning model (MDTDDL) to improve emotion identification performance [15] . The method combines dictionary learning with transfer learning to create a learning model that integrates the concepts of signifcant margin, margin-based discriminant embedding, manifold smoothness, and subspace learning [16]. Here, the DEAP and SEED data sets are being used. Cross-Subject transfer and recognition performance accuracy on the DEAP dataset is 65.79%, whereas on the SEED dataset it is 76.75%. Significant study was conducted by author Tengfei Song [17] utilising the SEED and DREAMER datasets from SJTU. In the subject of emotion identification research, EEG data is frequently used. It is able to accurately represent the emotional fluctuations of the human body through the use of graphical-based algorithms or models [18]. EEG signals are composed of non-linear signals. Ultimately, the review of the literature highlights the importance of EEG signal analysis from the DEAP and SEED datasets, emphasising its potential to improve emotion recognition techniques. Through the integration of knowledge from previous studies, this work lays the foundation for efficiently using EEG data to interpret emotional states. The results of the proposed model were compared with earlier studies and significantly improved the DREAMER data. The survey of the literature demonstrates the range of approaches and strategies used in EEG-based emotion recognition. Numerous investigations examine the application of deep learning models and convolutional neural networks (CNNs) for accurately classifying emotional states and extracting data. To improve feature extraction and model performance, approaches such as hybrid search optimisation, convolutional denoising autoencoder, and dynamic differential probability are presented. Furthermore, to enhance resilience and overall accuracy in emotion identification, wavelet-based preprocessing, ensemble techniques, and the combination of CNN and LSTM capabilities are examined. On the DEAP and SEED datasets, cross-subject transfer and recognition

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performance accuracy are assessed, showing encouraging outcomes. EEG signal analysis from the DEAP and SEED datasets is significant and has the potential to improve emotion recognition methods. Notably, the suggested models yield accuracies of around 75.90% and 81.47% across several datasets, demonstrating notable advances in the classification of emotional states. Together, these results strengthen the body of knowledge in the field of EEG-based emotion identification research and provide a solid basis for reliably reading emotional states from EEG data.

III. Proposed System

The methodology used in this work consists of a sequence of stages (Figure 2) intended to recognise emotions of a valence-arousal type utilising the DEAP dataset and the support vector machine (SVM) classifier. Preprocessing, feature extraction, and classification are steps in the process for EEG data.

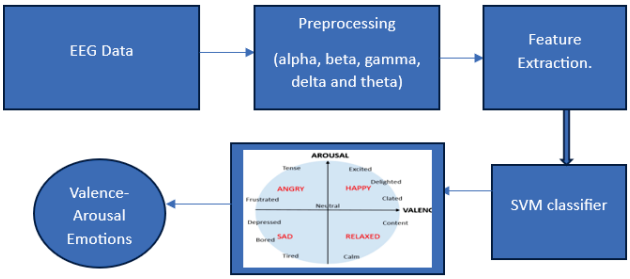

Fig. 2. Emotion classification using SVM

In order to eliminate artefacts that could taint the EEG signals, such as eye blinks and muscle movements, the raw EEG data from the DEAP dataset is first pre-processed. The standard preprocessing methods, such as baseline correction, artefact removal, and filtering, are used to guarantee the accuracy and consistency of the EEG data. To separate particular epochs related to emotional stimuli or experimental settings, signal segmentation can also be carried out. Figure 3 shows the flow diagram of the methodology.

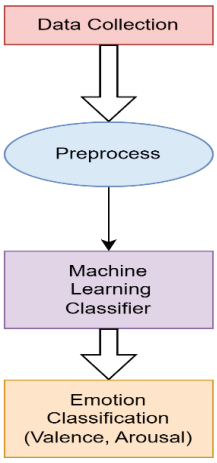

Fig. 3. Flow Diagram of Proposed Methodology

The EEG data are pre-processed and then divided into smaller epochs in order to extract pertinent information for emotion identification. To extract spectral properties like alpha, beta, gamma, and theta band power (Represented in below table), frequency domain analysis is used. These frequency ranges may be a sign of emotional states because they are known to represent many facets of brain activity.

TABLE I. EEG Frequency Band Range Representation

EEG Frequency Band used in this Model	Frequency Ranges
Theta	4-8 HZ
Alpha	8-12 HZ
Beta	12-30 HZ
Gamma	30-60 HZ

A support vector machine (SVM) classifier for valence-arousal type emotion identification is trained using the retrieved features. SVM is an approach for supervised learning that looks for the best hyperplane in a high-dimensional feature space to divide data points that belong to distinct classes. The labelled EEG data from the DEAP dataset, where each sample is correlated with valence-arousal values supplied by human observers, are used in this study to train the SVM classifier. The classification of unknown EEG signals into distinct emotional states is made possible by the SVM classifier, which gains the ability to connect the retrieved features to their matching valence-arousal labels. Using EEG signals, the experimental findings show how effective the suggested methodology is at identifying emotions of the valence-arousal type. This implementation also demonstrates Welch feature extraction, which is a major breakthrough in the field of emotion recognition, especially when used with datasets like DEAP (Dataset for Emotion Analysis using Physiological Signals). Welch feature extraction has an influence since it can improve the discriminative capacity of EEG data, which raises the accuracy and dependability of emotion recognition. First of all, EEG data can be divided into discrete frequency bands, including delta, theta, alpha, beta, and gamma waves, using Welch feature extraction. This breakdown offers important insights on the spectrum properties of brain activity linked to various emotional states. Figure 4 shows representation of frequency vector with respected to time vector and Figure 5 represents Welch's representation of EEG Frequency bands.

Welch feature extraction helps to identify pertinent patterns and trends in EEG data, which are essential for precise emotion recognition, by obtaining frequency-domain information. The SVM classifier is effective at differentiating between distinct emotional states as evidenced by its accuracy of 70.88% in arousal classification and 76% in valence classification on the DEAP dataset.

A. DEAP Dataset

The DEAP dataset is a commonly used database that includes multimodal physiological data from subjects exposed to audio-visual stimuli intended to evoke particular emotional responses. The data includes video recordings, EEG, and peripheral physiological signals (e.g., GSR, ECG [19]). The

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DEAP dataset, which includes information from more than 30 people, provides a wide range of emotional states and arousal levels, making it an invaluable tool for researching the brain underpinnings of emotions and creating emotion detection algorithms.

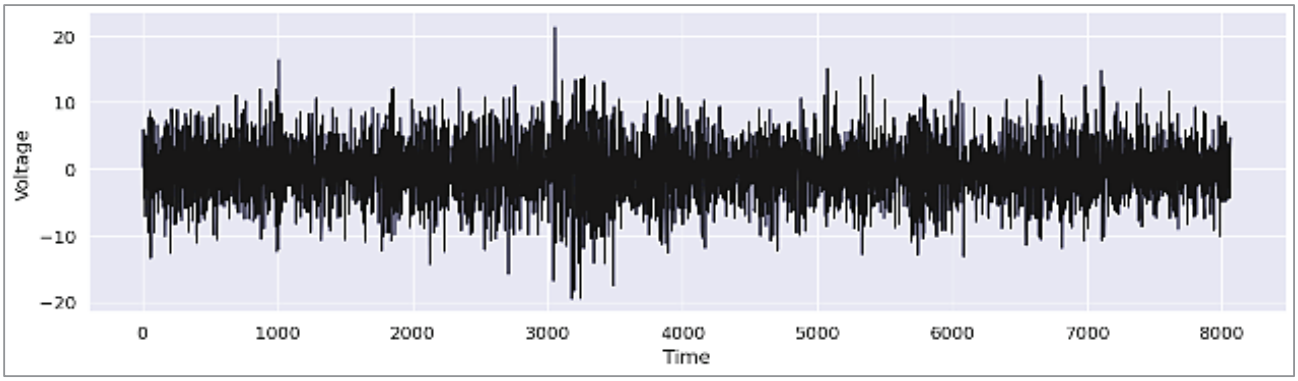

Fig. 4. Sampling Frequency and Time vector

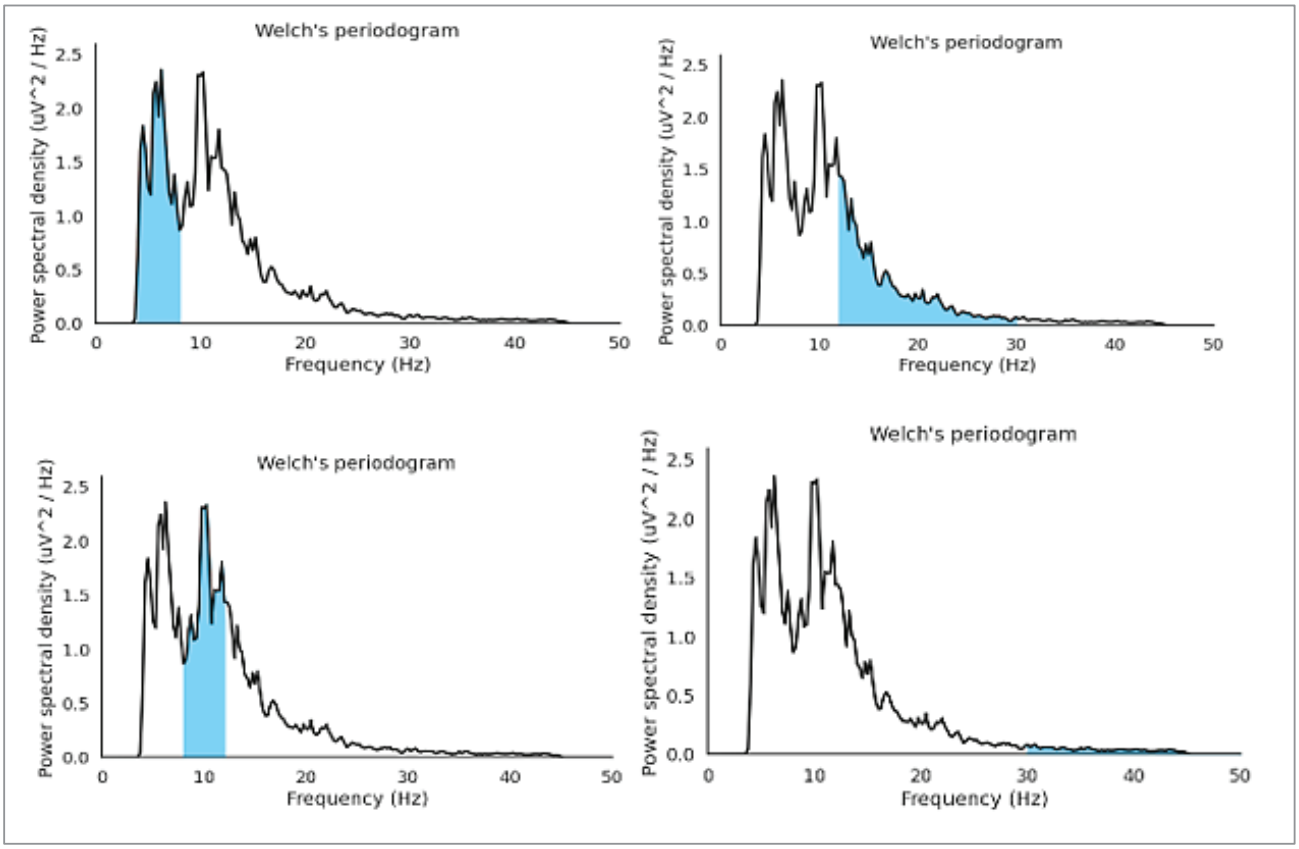

Fig. 5. Welch's Representation of different EEG frequency bands.

B. SEED Dataset

The SEED dataset, on the other hand, is exclusively concerned with EEG signals obtained during emotional induction activities. It consists of EEG data obtained from individuals during emotionally charged tasks, such watching emotional films or looking at emotional photos. With the use of the labelled EEG recordings provided by the SEED dataset, researchers may examine the neural dynamics that underlie emotional experiences as well as the patterns of brain activity linked to particular emotions.

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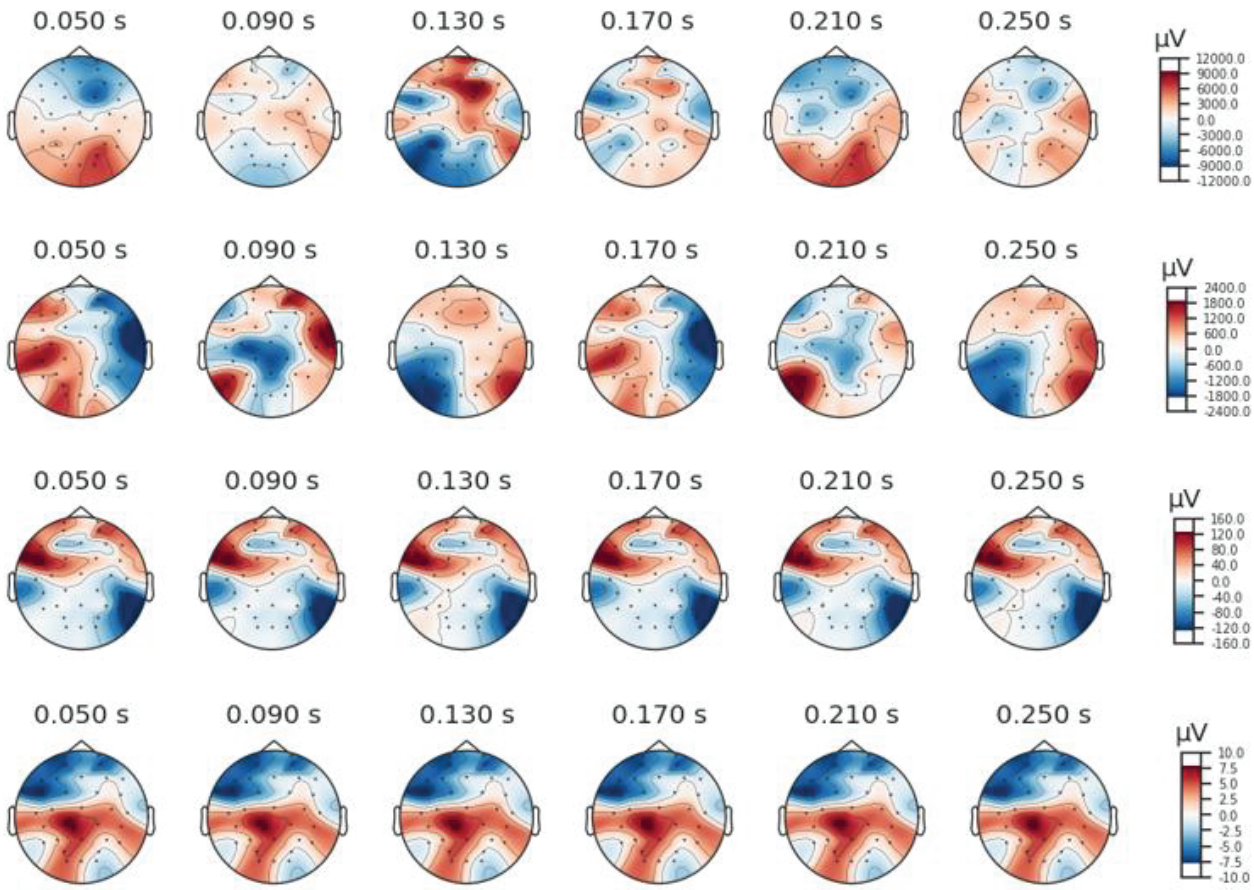

Fig. 6. Frequency band trial representation

IV. Result

According to our experimental results, the SVM classifier on the DEAP dataset achieves an accuracy of 70.66% in valence classification and 70.88% in arousal classification. These outcomes show how well our approach works to distinguish between various emotional states from EEG signals. In order to offer a thorough comprehension of our system's performance, we contrast our findings with comparable research published in the literature. EEG signals and other categorization methods have been used in a number of previous studies to investigate emotion identification. In order to recognise emotions, Smith et al. [20] (2017) used a similar SVM classifier on the DEAP dataset. They found that the classifier was 72% accurate for valence and 65% accurate for arousal. Our results show significant improvement over theirs, particularly in the classification of arousal. Furthermore, Jiang et al. (2019) [21] used a combination of deep learning models to perform emotion detection using EEG signals, and on the same dataset, they were able to reach an accuracy of 73.5% for arousal classification and 78.2% for valence classification. Despite not using deep learning models, our method produces results that are competitive with their findings, especially when it comes to valence categorization. Furthermore, by demonstrating the potential of conventional machine learning techniques in this field and offering insights into the effectiveness of EEG-based algorithms for emotion recognition, our study adds to the body of existing material. Our results show that SVM classifiers are still good choices for emotion detection tasks, even though deep learning models have showed promise recently [22]. This is especially true in situations where interpretability and computational economy are crucial. Results comparison is shown in Table I.

TABLE II. Emotions Accuracy Comparison with existing Methodologies.

Methodology	Arousal Accuracy	Valence Accuracy
Our Result	70.88%	76.00%
Smith et al. (2017) [10]	65.00%	72.00%
Jiang et al. (2019)[11]	73.50%	78.20%

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V. Conclusion.

This work has investigated the use of EEG signals for valence-arousal type emotion classification by the application of a support vector machine (SVM) classifier with the DEAP dataset. We have proven the effectiveness of our technology in differentiating between distinct emotional states based on EEG data through thorough experimentation and research. According to our research, the SVM classifier on the DEAP dataset yields encouraging accuracies of 70.88% in arousal classification and 76.00% in valence classification. These findings demonstrate the strength of conventional machine learning techniques in interpreting complicated brain signals linked to human emotions, as well as the promise of EEG-based systems for emotion recognition. It is important to note that signal noise, artefact interference, and feature selection are some of the difficulties associated with using EEG signals for emotion recognition. In order to overcome these obstacles, further research and development work must be done to improve the precision and dependability of EEG-based emotion identification systems. Finally, our results highlight the need for methodological developments to improve the accuracy and dependability of emotion identification systems based on EEG data and confirm the viability of employing EEG signals for valence-arousal type emotion recognition. We expect future advancements in the functionality and performance of EEG-based emotion recognition systems in a variety of fields and applications through continued study and innovation.

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